

Designing a Noise Reduction Filter for Voice Signals

A simulated public-announcement case study using frequency-domain analysis and a basic low-pass filter.

ECE0402 • Signals, Systems & Probability

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Why this problem matters

Real problem

Voice recordings and public announcements often lose clarity when background noise is present.

High-frequency hiss is a useful starting point because it is common in recording chains and easier to analyze.

Controllable model

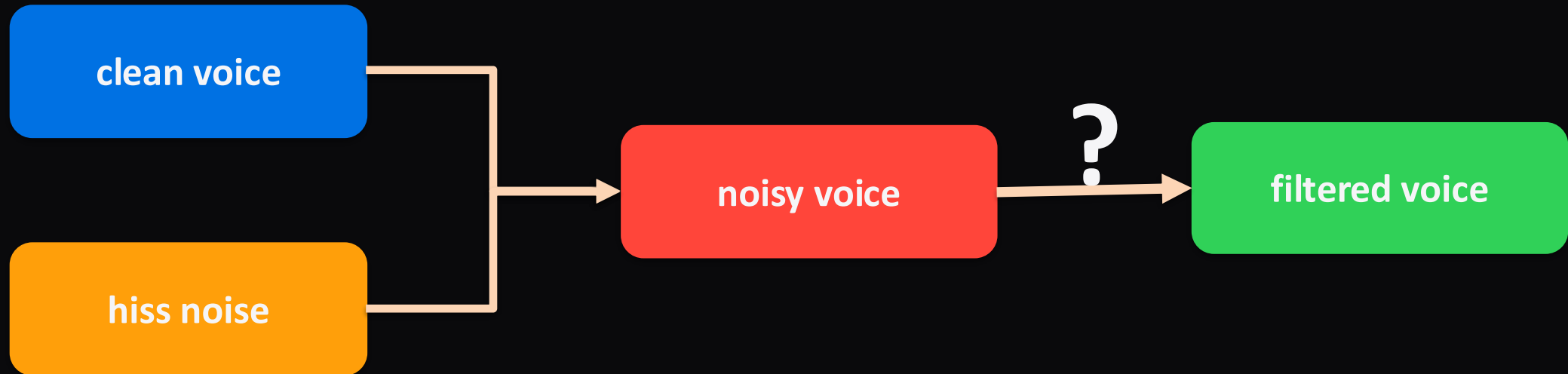
Explainable system

A simple, explainable filter is enough to show how Signals and Systems ideas work in a real application.

Problem statement

Input: noisy voice corrupted by high-frequency hiss.

Output: a cleaner voice signal after filtering.



MAIN DESIGN QUESTION

Where should the filter cut off to reduce noise without making speech sound too dull?

Simulated application scenario

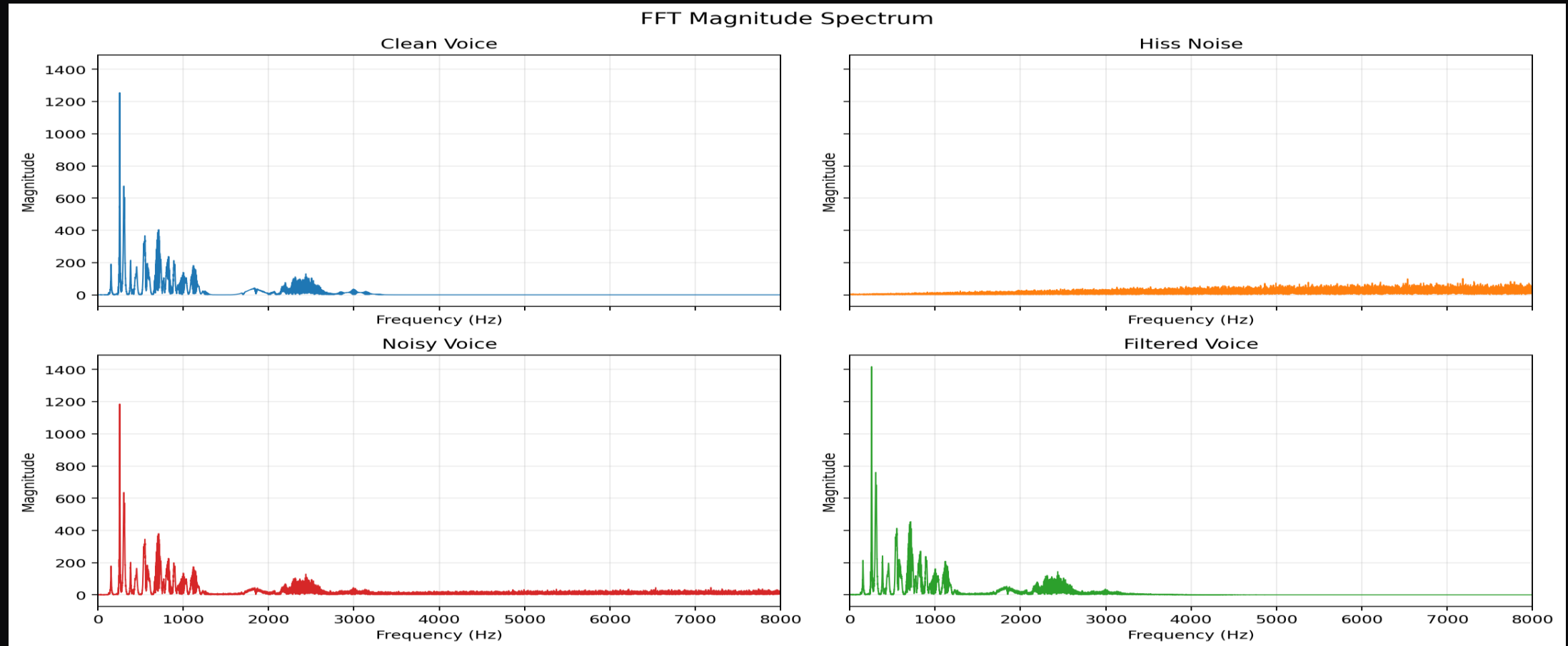
We use a station background as the story, but we simulate the signal instead of recording a complex real environment.

Scene framing

A short public-announcement style voice sample is corrupted by high-frequency hiss and then processed by a simple digital filter.



Signal characteristics

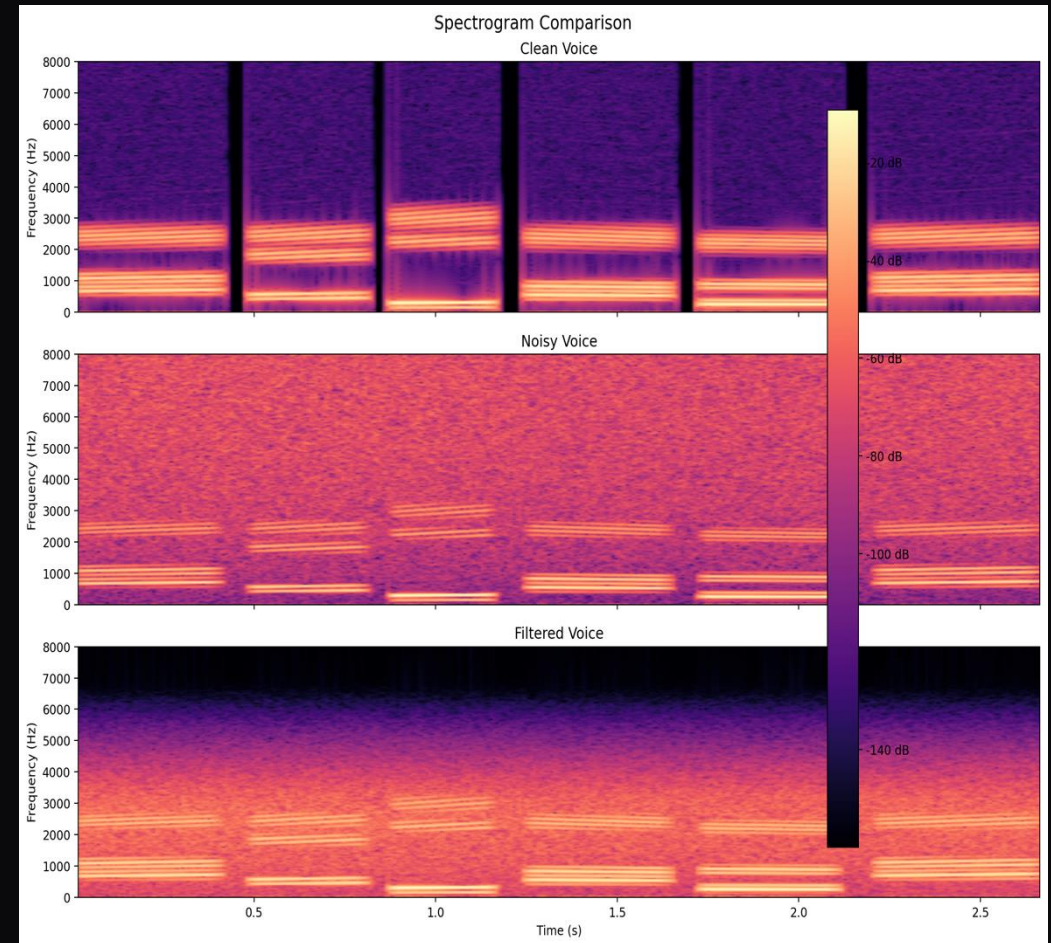


Why Fourier Transform helps

In the time domain, speech and noise are mixed together and hard to separate directly.

The Fourier Transform shows where energy is concentrated across frequencies.

That frequency view tells us why a low-pass filter is reasonable in this specific hiss scenario.

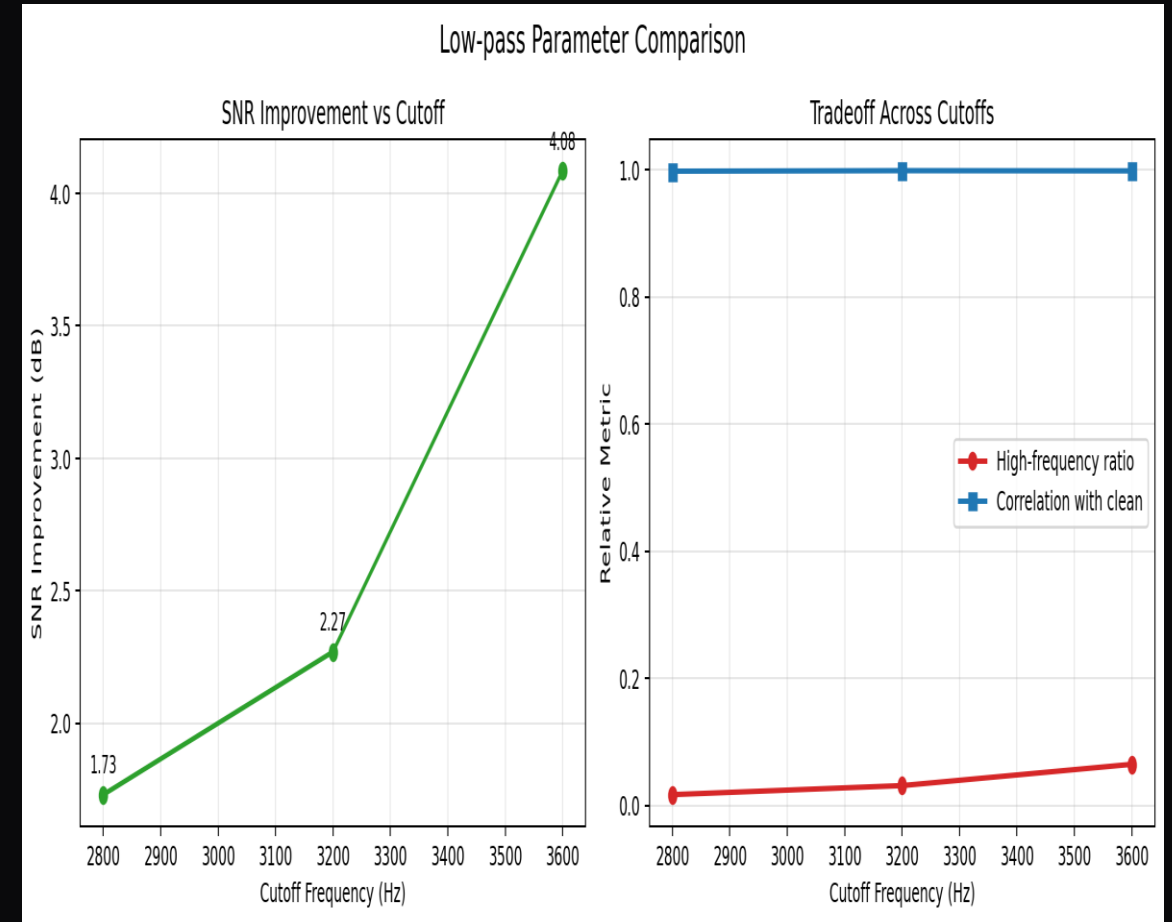


Filter design idea

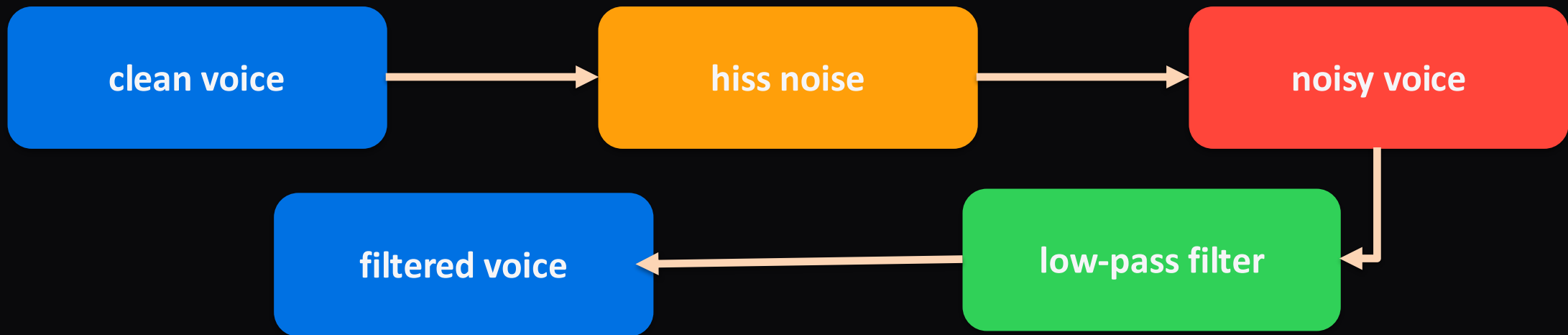
2800 Hz removes more hiss, but speech starts sounding dull.

3600 Hz preserves more speech detail, but leaves more high-frequency noise behind.

3200 Hz gives the best presentation balance between audible improvement and naturalness.



System workflow



Model a realistic but controllable scenario

Observe the signal in the frequency domain

Design and compare candidate filter parameters

Evaluate the result with audio and figures

Demo setup



Clean Voice

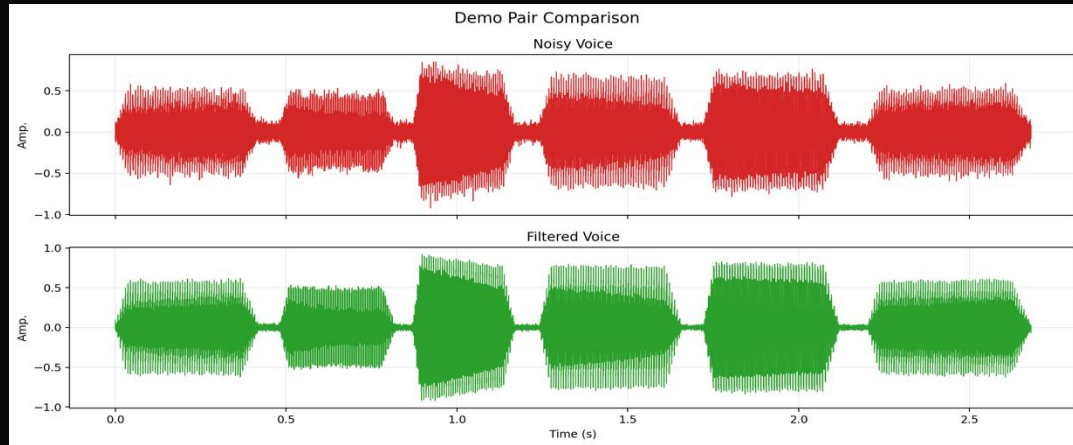


Noisy Voice



Filtered Voice

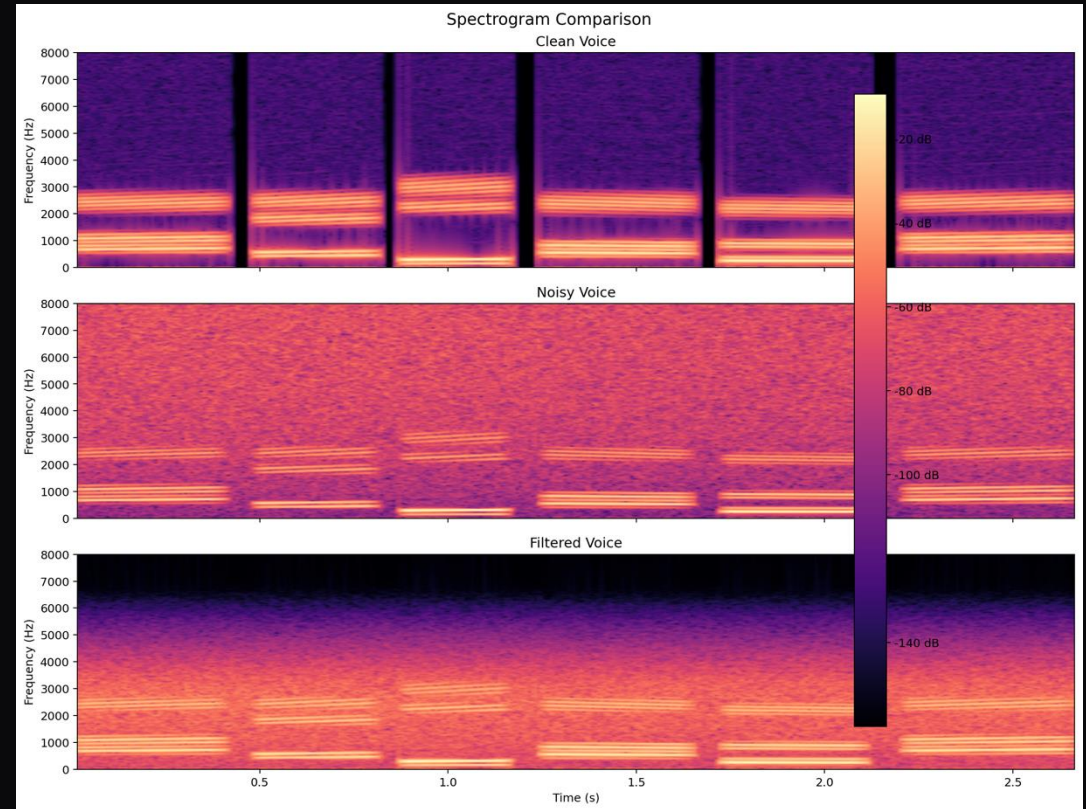
Results



High-frequency hiss is visibly reduced after filtering.

Speech remains understandable in the processed output.

The improvement is clear, but some high-frequency speech detail is softened.



Discussion and limitations

The low-pass filter works because the simulated hiss is mainly high-frequency.

The same mechanism also removes some useful high-frequency speech detail.

A more realistic station environment would contain broader and more complex noise, which basic low-pass filtering cannot handle as cleanly.

Suppression

Preservation

Tradeoff

A simple filter can explain a lot.

We modeled a voice-noise problem, used frequency-domain analysis to guide the design, and showed that a basic low-pass filter can reduce hiss while preserving intelligibility.